

Lab 2 : Importance of Cryptography

Main Objective

To examine how dummy login credentials are transmitted over HTTP and HTTPS using Wireshark, compare the captured packets, and explain how HTTPS encryption protects sensitive information from being accessed by attackers.

Tools Used

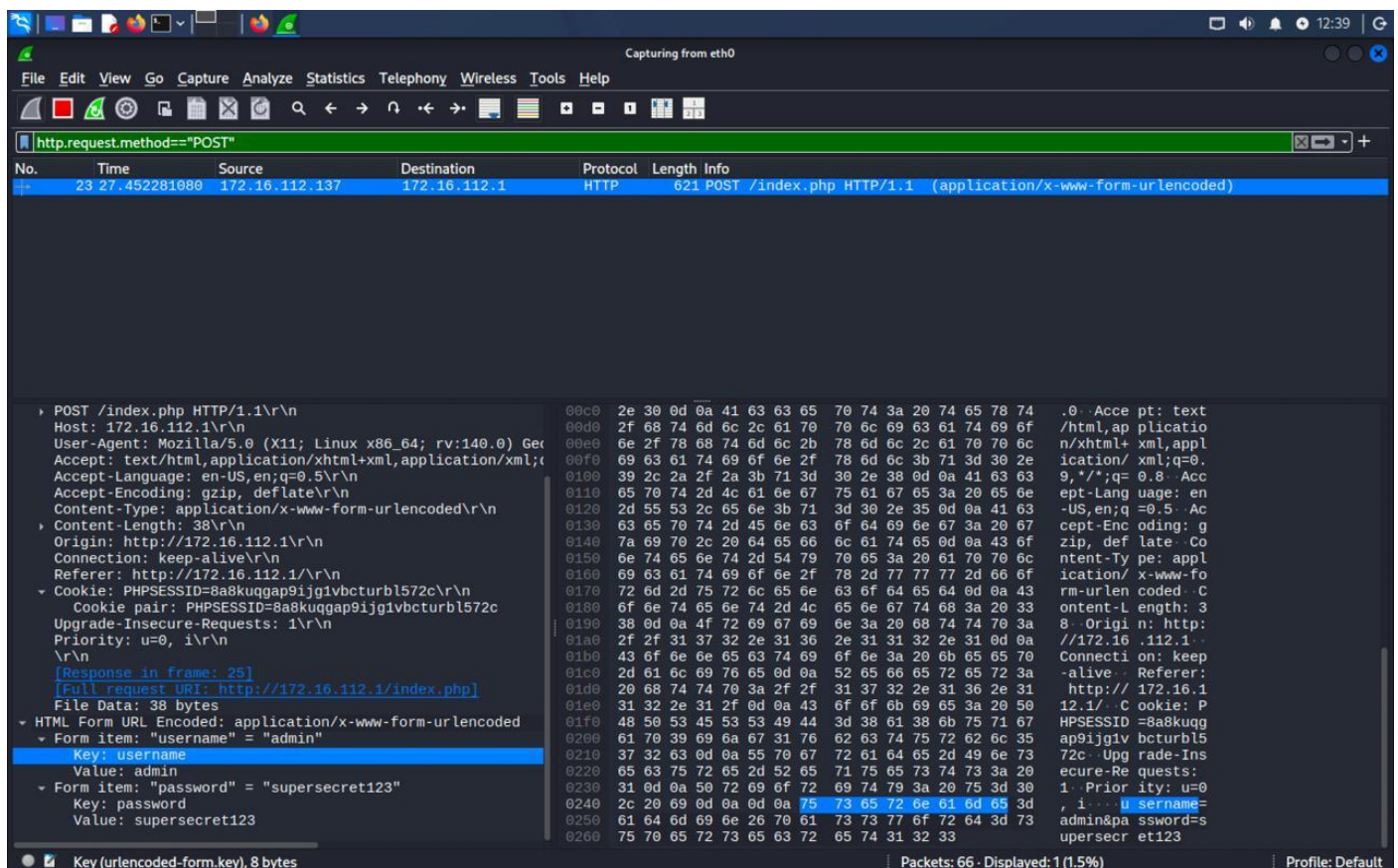
- Wireshark
- Web browser
- HTTP test login website
- GitHub login page

Procedure

Wireshark was used to capture the network traffic during two login experiments: one using an HTTP test login page and the other using an HTTPS-secured login page.

HTTP Login:

A packet capture was started on the active eth0 network interface before accessing an HTTP test login website. Dummy login credentials were entered and submitted, and the capture was then stopped. The captured HTTP POST request was analyzed, and the username and password were found to be visible in plain text within the HTML form data.



HTTPS Login:

A new packet capture was initiated on the same network interface before accessing the GitHub login page, which uses HTTPS. After entering and submitting the dummy login credentials, the capture was stopped and the packets were analyzed. Unlike HTTP, the login credentials were not visible in a readable format because the communication was protected by TLS encryption, which automatically encrypts the data during transmission.

The image shows a Wireshark packet capture of a TLSv1.3 connection. The top pane displays a list of packets, with packet 11 selected, showing a TLSv1.3 Application Data packet. The middle pane shows the packet details, including the TLSv1.3 Record Layer and the Application Data Protocol (Hypertext Transfer Protocol). The bottom pane shows the raw packet data in hexadecimal and ASCII. The status bar at the bottom indicates that the selected packet is encrypted application data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	172.16.112.137	172.16.112.1	TCP	74	41032 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2120829228 TSecr=26166782...
2	0.000564082	172.16.112.1	172.16.112.137	TCP	78	443 → 41032 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=64 TSval=26166782...
3	0.000602904	172.16.112.137	172.16.112.1	TCP	66	41032 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2120829221 TSecr=26166782...
4	0.000752526	172.16.112.1	172.16.112.137	TCP	66	[TCP Window Update] 443 → 41032 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=26166782...
5	0.002200596	172.16.112.137	172.16.112.1	TLSv1.3	2465	Client Hello
6	0.002375127	172.16.112.1	172.16.112.137	TCP	66	443 → 41032 [ACK] Seq=1 Ack=2400 Win=129472 Len=0 TSval=2616678205 TSecr=2120...
7	0.010935708	172.16.112.1	172.16.112.137	TLSv1.3	306	Server Hello, Change Cipher Spec, Application Data, Application Data
8	0.010970198	172.16.112.137	172.16.112.1	TCP	66	41032 → 443 [ACK] Seq=2400 Ack=241 Win=64512 Len=0 TSval=2120829231 TSecr=261...
9	0.011549734	172.16.112.137	172.16.112.1	TLSv1.3	130	Change Cipher Spec, Application Data
10	0.011669448	172.16.112.1	172.16.112.137	TCP	66	443 → 41032 [ACK] Seq=241 Ack=2464 Win=131008 Len=0 TSval=2616678214 TSecr=21...
11	0.011813989	172.16.112.137	172.16.112.1	TLSv1.3	765	Application Data
12	0.011944242	172.16.112.1	172.16.112.137	TCP	66	443 → 41032 [ACK] Seq=241 Ack=3163 Win=130432 Len=0 TSval=2616678215 TSecr=21...
13	0.012370032	172.16.112.1	172.16.112.137	TLSv1.3	337	Application Data

[TCP Segment Len: 699]
Sequence Number: 2464 (relative sequence number)
Sequence Number (raw): 1015926142
[Next Sequence Number: 3163 (relative sequence number)
Acknowledgment Number: 241 (relative ack number)
Acknowledgment number (raw): 724177885
1000 = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 126
[Calculated window size: 64512]
[Window size scaling factor: 512]
Checksum: 0x3b8d [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP)
TCP Option - No-Operation (NOP)
TCP Option - No-Operation (NOP)
TCP Option - Timestamps: TSval 2120829232, TSecr 2616678205
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (699 bytes)
Transport Layer Security
[Stream index: 0]
TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

Bytes 71-764: Encrypted Application Data (tls.app_data) | Packets: 42 | Profile: Default

The image shows a web browser window with the address bar displaying "172.16.112.1/index.php". The page title is "Employee Portal Login". The page content includes a login form with fields for "Username" and "Password", and a "Login" button. Below the form, there is a "Lab Goal" section that reads: "Capture the POST request via HTTP to find the password, then try via HTTPS to see the encryption." The browser's address bar shows the URL "172.16.112.1/index.php".

Employee Portal

Login

Lab Goal: Capture the POST request via HTTP to find the password, then try via HTTPS to see the encryption.



Comparison

Features	HTTP	HTTPS
Data Transmission	Data is transmitted without encryption, making login information potentially visible in Wireshark.	Data is encrypted using TLS, so the actual login information cannot be directly read.
Security & Confidentiality	Does not ensure confidentiality, leaving sensitive information vulnerable to interception.	Provides confidentiality and enhanced protection for sensitive information during transmission.
Use for Login	Not appropriate for securely transmitting usernames and passwords.	Recommended for login pages and other communications involving sensitive information.

Importance of Cryptography

Cryptography helps protect sensitive information when it is transmitted over a network. In this experiment, the login credentials sent through HTTP were directly visible in Wireshark because the data was not encrypted. When HTTPS was used, the login information was protected by TLS encryption, making it impossible to read directly from the captured packets. This demonstrates how cryptography prevents sensitive information from being easily accessed by someone monitoring the network. It also helps maintain the confidentiality and integrity of data during transmission.

Conclusion

This experiment helped us understand the practical difference between HTTP and HTTPS. In the HTTP capture, the dummy login credentials were visible in Wireshark, whereas in the HTTPS capture, the credentials were encrypted and could not be read directly. This demonstrates how encryption helps protect sensitive information, such as usernames and passwords, when transmitted over a network.